What steers an LLM's next experiment? Results, names, and wording in reaction optimization

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Preprint. Work in progress. One model, measured chemistry data, no human participants.

Abstract

Bayesian optimization (BO) has outperformed expert chemists at reaction optimization, and scientists now increasingly ask language models what to try next. Whether that brings BO-quality decisions is an open question. A second question concerns direction. Language models reflect what has already been reported, so a person working with one may be drawn toward canonical answers, which is a risk when those answers are wrong, thinly supported, or slightly beside the point. Someone who knows less may sometimes search more freely, a possibility we raise here and do not test. Before testing either question, we measured what steers a model's suggestion. On 160 histories from measured reaction grids, we asked Claude Opus 5.5 for the next three experiments, then changed one input at a time. Repeating the request changed the picks by 0.15 on a 0 to 1 scale. A different prompt changed them by 0.17, advice by 0.24, a hunch by 0.26, coded or swapped names by 0.41 and 0.46, and scrambled results by 0.46. The influence of results grew with the amount of data, and the influence of names declined. Advice to rely on the data shifted picks toward the results shown without improving their true rank. All of these results describe one model, and many of the questions would be better tested with experts and novices in these fields. We did not test BO itself or any effect on human reasoning, and propose both as next steps.

1. The claim

Experimental optimization is often done by intuition and by changing one factor at a time. Shields et al. [1] put human judgment directly against BO. Fifty chemists and engineers played an online game on a fully measured direct arylation grid, choosing five experiments per round, and BO outperformed them in efficiency and consistency. In the paper, BO started at random. Players made better first choices, but BO's average passed theirs within three rounds, and the difference on the recorded data was significant after the fifth. The released data also include BO runs started from the players' own first choices. In those runs [2], by 25 experiments experts averaged a best yield of 91.7% and BO 96.7%, and 50% of experts versus 86% of BO runs had reached 90% (Figure 1). If trained experts have something to learn from BO, most practitioners have more.

BO is hard to adopt, however. It needs code, statistical judgment, and setup. Many researchers will instead paste their results into a chat and ask what to try next.

2. The question

Much problem solving interpolates between known results, and language models are strong at this because they are trained on what has already been reported. Discovery sometimes depends on the opposite: a result outside that distribution, where the canonical answer is wrong, rests on thin evidence, or slightly misses the point. A tool that steers people toward

the canonical answer could help in familiar territory and hurt at the frontier. Paradigm shifts have often come from outside the accepted picture, which raises the possibility that knowing less, or reading less of the existing work first, sometimes leads to a better search. We state this as a hypothesis.

One model-side result points the same way. When Cissé et al. [8] attached a 2020 paper on the same reaction to a GPT-5/BO optimizer, the average best rate at 25 experiments rose from 7.5 to 18.5, but at 150 experiments it fell from 25.1 to 21.8, and the optimum was reached in 0 of 20 runs versus 9 of 20. The authors speculate that the paper's deselection of salt steered the search away from a benefit found later. Attaching the underlying data instead raised the final result to 26.6, and letting the model search the literature itself made little net difference. This is one simulated problem with one attached paper, so it is an example and does not establish the hypothesis.

How do language models affect this kind of decision? When do they enable better search, and when do they pull a person toward what is already believed? A model's suggestion can draw on the results it is shown, on knowledge from training, and on how the question is phrased. We start by measuring the relative influence of each, which covers the model's side of that interaction only.

3. Prior work and the gap

Work	Conclusion	Gap left
Shields 2021 [1]	BO beat experts on one reaction grid	No LLMs
Gupta 2025 [3]	LLM agents barely reacted to shuffled feedback on gene tasks, and classical methods beat them	Older models. No test of names or wording
MacKnight 2025 [4]	LLMs matched or exceeded BO on five single-objective reaction grids. BO kept the advantage on the one multi-objective case	Credits prior knowledge, with no condition that removes it
Müller 2026 [5]	Agents can run BO campaigns on real platforms. Scripted loops beat step-by-step tool calls	All arms already use BO, so plain chat is not compared
Cooper 2026 [6]	An expert team's best catalyst matched the stronger LLM run within error and beat the weaker one, on one problem. The final best was nearly absent from LLM proposals made without feedback	One problem, no test of inputs
Cissé 2026 [8]	On two simulated problems, LLM and LLM/BO search beat BO. Bad hypotheses lowered performance, and most runs at least partly recovered from data	Noise-free simulators. Inputs tested on one problem and scored by search outcome. Names stay real and results are not scrambled
Zhang 2026 [7]	Experimental design was the most sycophancy-prone task studied	Single-turn, scored on text

Cissé et al. [8] come closest, varying the hypotheses, literature, and data given to a model. We did not find a controlled measurement of how results, names, and wording each move an

LLM's choice of next experiment. This claim covers the papers listed here and the main text of [8], whose supplement we did not read.

4. What we tested

We used complete measured grids with open licenses: direct arylation (1,728 yields) [1], a Suzuki coupling table (3,696 reactions) from the EDBO repository [9], oil palm delignification (64 conditions in duplicate) [10], and blocks of a C–N coupling screen [11], of which the first 10 of 29 eligible blocks were used. Each of the four sources contributed equally.

One unit is a world, a random history of 0, 3, 6, or 12 measured results, and a prompt from a pool of 40 varied templates. Each unit was sent to Opus 5.5 once per condition (Figure 2), with no tools, 160 units and 1,720 decisions in all. Each comparison changes one input. Sampling and analysis were fixed before this run, after earlier exploratory work on the design.

We report change in picks (total-variation distance between the option distributions of two sets of three picks, averaged over settings), true rank (grid percentile of the picks), and agreement with the best option in the results shown and with the model's own no-results ranking.

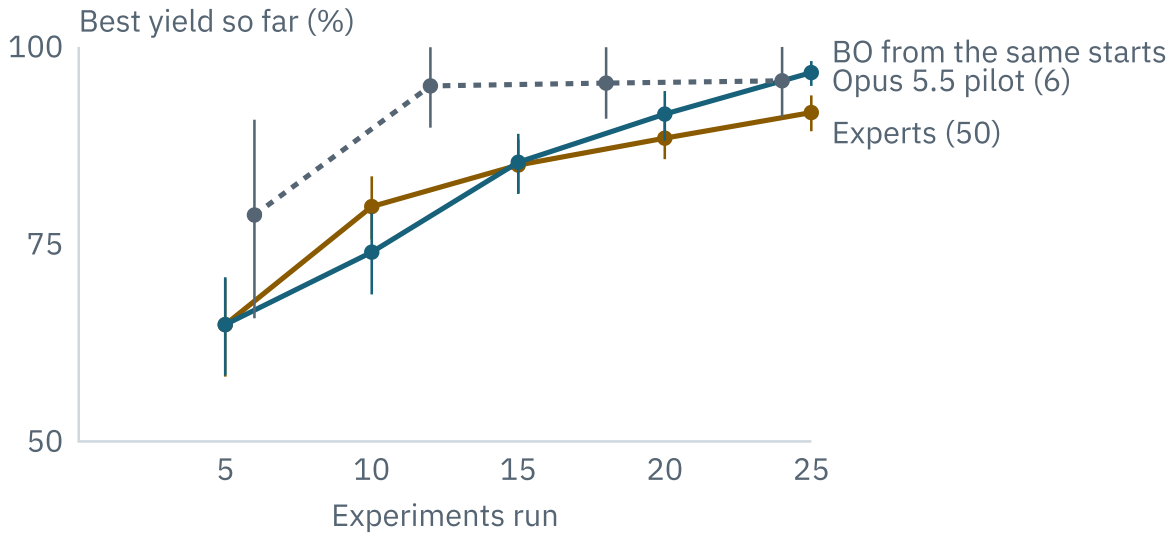


Figure 1. Best yield so far on the direct arylation grid. Expert and BO curves use the released game data [2]. These BO runs started from each player's first batch, whereas the paper's own comparison [1] started BO at random. Players could stop early and keep their best result, which matches the paper's lower bound on human performance. The Opus points are a 6-run exploratory pilot in batches of 3, on a grid the model named as a known dataset, so they are illustrative. Intervals are 95%.

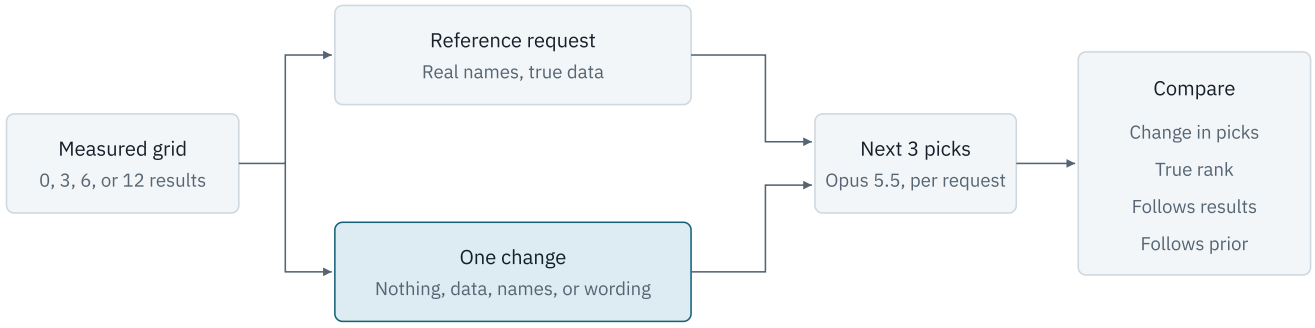


Figure 2. One unit of the replay study. Because the history is shared, each comparison isolates a single input. Repeating the reference request gives a noise floor.

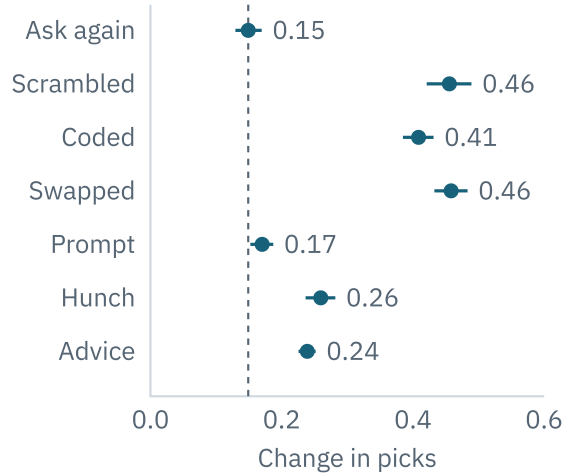


Figure 3. Change in picks by input, with 95% intervals. The dashed line is the noise floor.

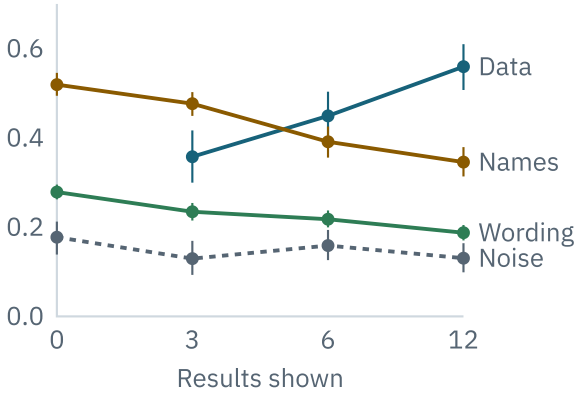


Figure 4. Change in picks against the number of results shown. Names pools coded and swapped. Wording pools prompt, hunch, and advice.

5. Results

All 1,720 decisions were usable. Repeating the same request changed the picks by 0.15. A different prompt changed them by 0.17, which exceeds the noise floor by 0.02 (interval 0.00 to 0.04). Advice, hunches, names, and scrambled results each changed picks by more (Figure 3).

Scrambling mattered more as data accumulated, rising from 0.36 at 3 results to 0.56 at 12. Names mattered less, falling from 0.52 with no results to 0.35 at 12 (Figure 4). Sources differed: on the C–N blocks, names moved picks about twice as far as scrambling, the reverse of the other three (Table 1).

Evidence-first and audit advice raised agreement with the best observed option and lowered agreement with the model's prior, but true rank fell by about 0.02 (Table 2).

The model's no-results rankings tracked measured option quality more often than not. Mean Spearman correlation with measured option means was 0.72 on direct arylation, 0.64 on Suzuki, and 0.38 on the C–N blocks, and its top choice matched the best option in 60%, 60%, and 27% of settings. On oil palm, its ranking of reaction time ran opposite to the measurements (−0.52). It named the source of direct arylation and Suzuki data correctly in every try.

Table 1. Mean change in picks by input and source.

Input	Shields	Suzuki	Oil palm	Cernak
Ask again	0.14	0.13	0.16	0.17
Scrambled	0.47	0.59	0.48	0.29
Coded	0.35	0.35	0.26	0.67
Swapped	0.45	0.34	0.35	0.70
Prompt	0.15	0.15	0.18	0.20
Hunch	0.27	0.21	0.21	0.34
Advice	0.24	0.19	0.20	0.33

Table 2. Paired difference from the reference request, with 95% intervals.

Strategy	True rank	Follows results	Follows prior
Evidence first	−0.02 [−0.04, −0.01]	0.04 [0.02, 0.07]	−0.06 [−0.08, −0.05]
Audit assumptions	−0.02 [−0.04, −0.01]	0.09 [0.06, 0.12]	−0.08 [−0.09, −0.07]
Caution	−0.01 [−0.03, 0.01]	0.03 [0.00, 0.05]	−0.07 [−0.09, −0.06]
No rationale	−0.00 [−0.02, 0.01]	−0.05 [−0.07, −0.03]	0.01 [−0.01, 0.02]

6. Significance

These results suggest three practical points, each to be confirmed against outcomes. Rewording a request changed little beyond ordinary repetition. A user's results strongly influenced picks once several were available, so early suggestions lean more on the model's learned associations than later ones. Telling a model to rely on the data changed its choices but did not improve them here, so that advice alone is unlikely to substitute for an actual optimizer.

For model developers, shuffled-feedback and mislabeled-option tests are cheap checks of whether a model updates on evidence. For scientists, the evidence does not yet settle a default. BO beat experts in Shields [1], while LLM or LLM/BO search matched or beat BO in [4] and, on simulated noise-free problems, in [8]. The head-to-head below would test this on measured grids.

7. Limits and next steps

This study used one model, and chemistry data, two of whose grids the model recognizes. Replay measures single decisions on random histories. It does not measure a sequential optimization loop or what people do. The Figure 1 comparison spans different batch sizes and sample sizes.

Most of these questions would be better answered with people. Whether a suggestion steers reasoning, helps a novice, or misleads an expert can only be measured by giving suggestions to novices and experts in the field and observing what they do. We did not have access to such participants, so Claude Opus 5.5 is the only subject of the replay study. Its results describe one model's behavior and are not estimates of how scientists reason. The practical points in Section 6 are suggestions to test with practitioners.

Three studies follow directly. First, a head-to-head search on these grids: plain chat asked for help, chat asked for BO, the model with code execution, and the model with a real BO tool, scored against BO alone. Cissé et al. [8] compared LLM-only, hybrid, and BO search on simulators, and this would add measured grids and the plain-chat and asked-for-BO arms. Second, simulations validated against measured data, which would allow unlimited sequential experiments in settings the model has not seen. Third, a study with people guided in these different ways, using the Shields game as a template, since it is the only design here able to test whether model suggestions steer human reasoning.

Data and code

Grid sources, the prompt pool, the unit table, and analysis code are listed in `data/SOURCES.md` and the project repository.

Every request and response is logged.

References

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